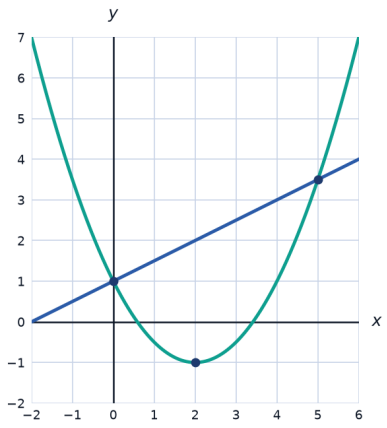
Quadratics & Complex Numbers

ClipSAT Printable Practice Suite (clipsat.org) • Topic 2.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Factoring polynomials completely

For the model $f(x) = 4x + 8$ in factoring polynomials completely, evaluate $f(5)$.

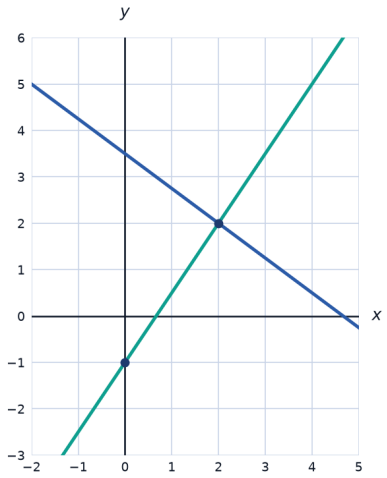


2 [Foundational] • Target: Factoring polynomials completely

Find the positive solutions to the quadratic equation $x^2 - 10x + 24 = 0$.

3 [Foundational] • Target: Review: Graphs and characteristics of quadratic functions

For the model $f(x) = 5x + 9$ in review: graphs and characteristics of quadratic functions, evaluate $f(2)$.



4 [Foundational] • Target: Review: Graphs and characteristics of quadratic functions

Find the positive solutions to the quadratic equation $x^2 - 5x + 4 = 0$.

5 [Foundational] • Target: Review: Graphs and characteristics of quadratic functions

A data set recorded for review: graphs and characteristics of quadratic functions consists of the four values $\{14, 18, 22, 28\}$. What is the arithmetic mean? Show your complete algebraic work.

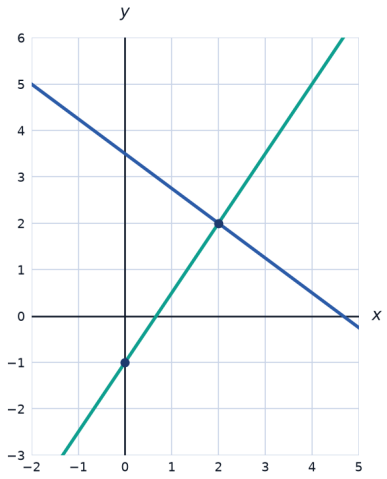
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6 [Foundational] • Target: Review: Graphs and characteristics of quadratic functions

For the model $f(x) = 5x + 9$ in review: graphs and characteristics of quadratic functions, evaluate $f(3)$.

7 [Foundational] • Target: Review: Graphs and characteristics of quadratic functions

Find the positive solutions to the quadratic equation $x^2 - 7x + 10 = 0$.



Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Review: Graphs and characteristics of quadratic functions

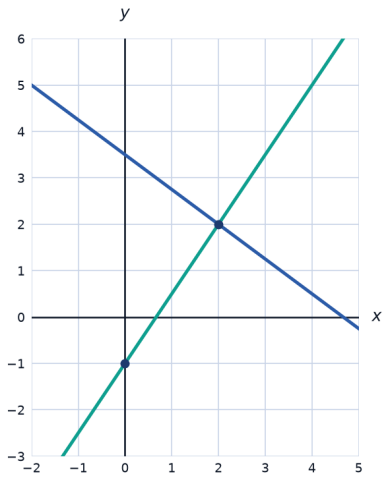
For the model $f(x) = 5x + 9$ in review: graphs and characteristics of quadratic functions, evaluate $f(4)$.

9 [Intermediate] • Target: Review: Graphs and characteristics of quadratic functions

Find the positive solutions to the quadratic equation $x^2 - 9x + 18 = 0$.

10 [Intermediate] • Target: Review: Graphs and characteristics of quadratic functions

For the model $f(x) = 5x + 9$ in review: graphs and characteristics of quadratic functions, evaluate $f(5)$.



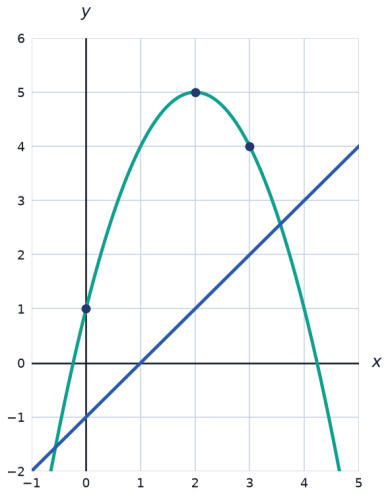
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11 [Intermediate] • Target: Review: Graphs and characteristics of quadratic functions

Find the positive solutions to the quadratic equation $x^2 - 11x + 28 = 0$.

12 [Intermediate] • Target: Review: Solving quadratic equations by factoring

For the model $f(x) = 6x + 3$ in review: solving quadratic equations by factoring, evaluate $f(2)$.



13 [Intermediate] • Target: Review: Solving quadratic equations by factoring

A data set recorded for review: solving quadratic equations by factoring consists of the four values $\{15, 19, 23, 29\}$. What is the arithmetic mean? Show your complete algebraic work.

14 [Intermediate] • Target: Review: Solving quadratic equations by factoring

For the model $f(x) = 6x + 3$ in review: solving quadratic equations by factoring, evaluate $f(3)$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

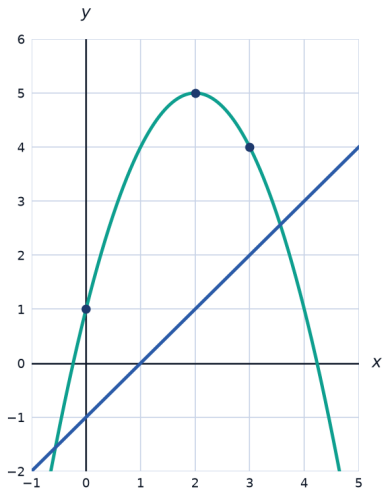
15 [Challenging] • Target: Review: Solving quadratic equations by factoring • Cross-Lesson: 7.01 Sequences & Series

For the model $f(x) = 6x + 3$ in review: solving quadratic equations by factoring, evaluate $f(4)$.

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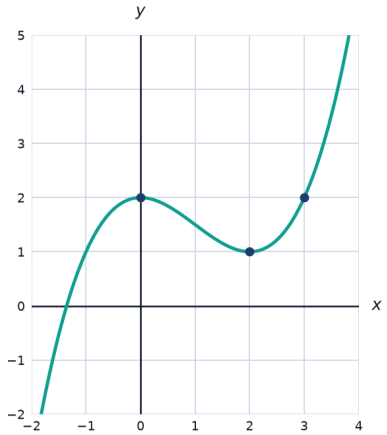
16 [Challenging] • Target: Review: Solving quadratic equations by factoring • Cross-Lesson: 8.01 Conic Sections

For the model $f(x) = 6x + 3$ in review: solving quadratic equations by factoring, evaluate $f(5)$.



17 [Challenging] • Target: Complex numbers • Cross-Lesson: 9.01 Trigonometric Functions

For the model $f(x) = 2x + 4$ in complex numbers, evaluate $f(2)$.



18 [Challenging] • Target: Complex numbers • Cross-Lesson: 1.01 Functions & Their Graphs

A data set recorded for complex numbers consists of the four values $\{16, 20, 24, 30\}$. What is the arithmetic mean? Show your complete algebraic work.

19 [Challenging] • Target: Complex numbers • Cross-Lesson: 10.01 Matrices & Systems

For the model $f(x) = 2x + 4$ in complex numbers, evaluate $f(3)$.

20 [Challenging] • Target: Complex numbers • Cross-Lesson: 11.01 Probability & Statistics

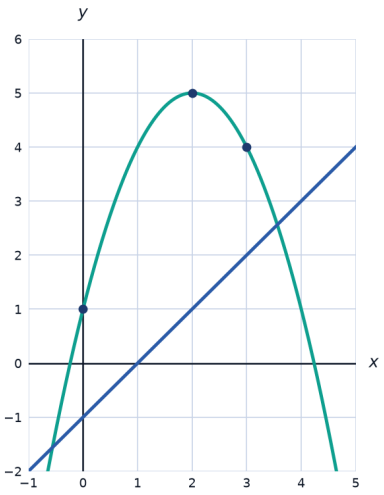
For the model $f(x) = 2x + 4$ in complex numbers, evaluate $f(4)$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Review: Factoring techniques

For the model $f(x) = 2x + 6$ in review: factoring techniques, evaluate $f(2)$.



a. 10

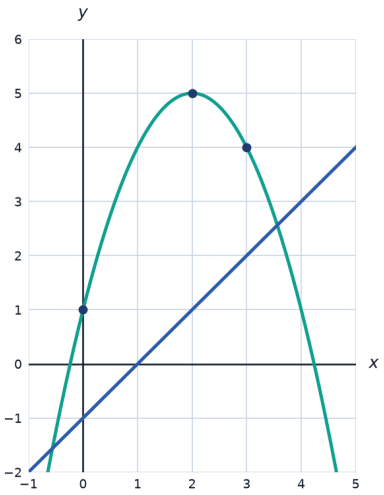
b. 12

c. 4

d. 14

22 [Foundational] • Target: Review: Factoring techniques

Find the positive solutions to the quadratic equation $x^2 - 8x + 12 = 0$.



a. $x = 2$ and $x = 7$

b. $x = 2$ and $x = 6$

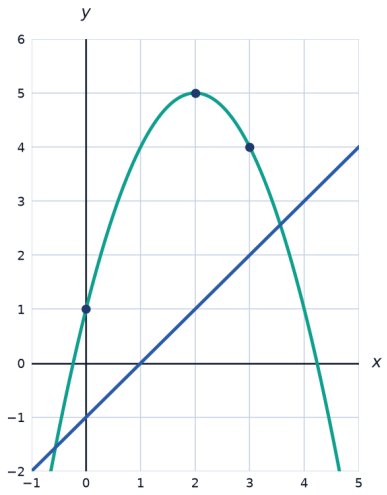
c. $x = -2$ and $x = -6$

d. $x = 8$ and $x = 12$

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23 [Foundational] • Target: Review: Factoring techniques

For the model $f(x) = 2x + 6$ in review: factoring techniques, evaluate $f(5)$.



a. 16

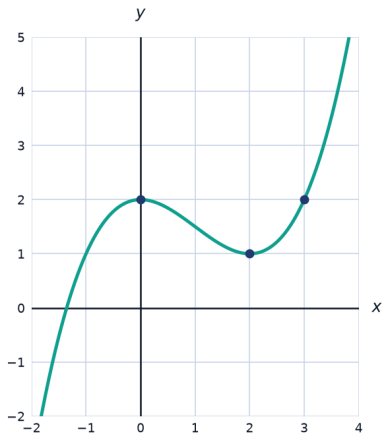
b. 18

c. 10

d. 20

24 [Foundational] • Target: Factoring with sum and difference of cubes

For the model $f(x) = 3x + 7$ in factoring with sum and difference of cubes, evaluate $f(2)$.



a. 13

b. 16

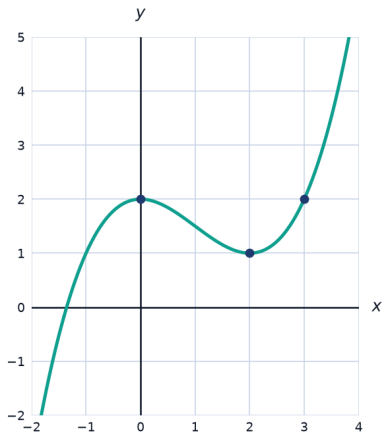
c. 6

d. 17

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25 [Foundational] • Target: Factoring with sum and difference of cubes

Find the positive solutions to the quadratic equation $x^2 - 9x + 14 = 0$.



a. $x = 2$ and $x = 8$

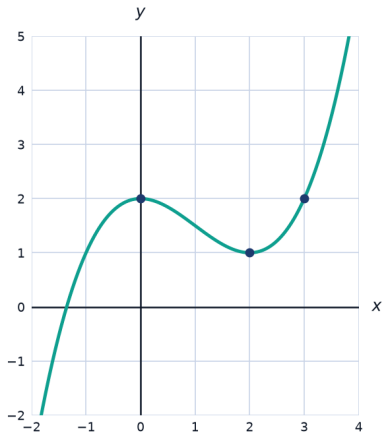
b. $x = 2$ and $x = 7$

c. $x = -2$ and $x = -7$

d. $x = 9$ and $x = 14$

26 [Foundational] • Target: Factoring with sum and difference of cubes

For the model $f(x) = 3x + 7$ in factoring with sum and difference of cubes, evaluate $f(5)$.



a. 22

b. 25

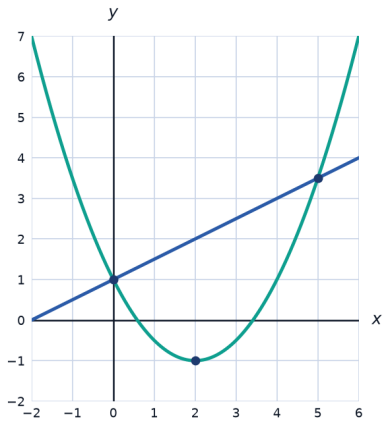
c. 15

d. 26

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27 [Foundational] • Target: Factoring polynomials completely

For the model $f(x) = 4x + 8$ in factoring polynomials completely, evaluate $f(2)$.



a. 16

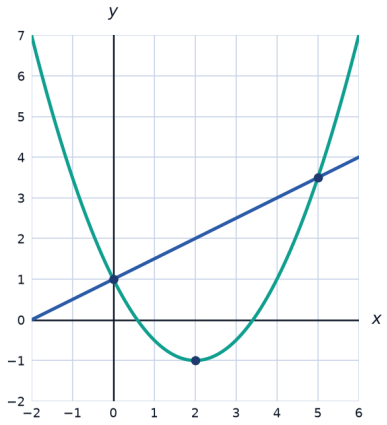
b. 20

c. 8

d. 20

28 [Foundational] • Target: Factoring polynomials completely

Find the positive solutions to the quadratic equation $x^2 - 6x + 8 = 0$.



a. $x = 2$ and $x = 5$

b. $x = 2$ and $x = 4$

c. $x = -2$ and $x = -4$

d. $x = 6$ and $x = 8$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Review: Factoring techniques

Find the positive solutions to the quadratic equation $x^2 - 6x + 5 = 0$.

a. $x = 1$ and $x = 6$

b. $x = 1$ and $x = 5$

c. $x = -1$ and $x = -5$

d. $x = 6$ and $x = 5$

30 [Intermediate] • Target: Review: Factoring techniques

A data set recorded for review: factoring techniques consists of the four values $\{11, 15, 19, 25\}$. What is the arithmetic mean?

a. 15.5

b. 16.5

c. 17.5

d. 19.5

31 [Intermediate] • Target: Review: Factoring techniques

For the model $f(x) = 2x + 6$ in review: factoring techniques, evaluate $f(3)$.

a. 12

b. 14

c. 6

d. 16

32 [Intermediate] • Target: Review: Factoring techniques

For the model $f(x) = 2x + 6$ in review: factoring techniques, evaluate $f(4)$.

a. 14

b. 16

c. 8

d. 18

33 [Intermediate] • Target: Review: Factoring techniques

Find the positive solutions to the quadratic equation $x^2 - 10x + 21 = 0$.

a. $x = 3$ and $x = 8$

b. $x = 3$ and $x = 7$

c. $x = -3$ and $x = -7$

d. $x = 10$ and $x = 21$

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34 [Intermediate] • Target: Review: Factoring techniques

Find the positive solutions to the quadratic equation $x^2 - 12x + 32 = 0$.

a. $x = 4$ and $x = 9$

b. $x = 4$ and $x = 8$

c. $x = -4$ and $x = -8$

d. $x = 12$ and $x = 32$

35 [Intermediate] • Target: Factoring with sum and difference of cubes

A data set recorded for factoring with sum and difference of cubes consists of the four values $\{12, 16, 20, 26\}$. What is the arithmetic mean?

a. 16.5

b. 17.5

c. 18.5

d. 20.5

36 [Intermediate] • Target: Factoring with sum and difference of cubes

For the model $f(x) = 3x + 7$ in factoring with sum and difference of cubes, evaluate $f(3)$.

a. 16

b. 19

c. 9

d. 20

37 [Intermediate] • Target: Factoring with sum and difference of cubes

For the model $f(x) = 3x + 7$ in factoring with sum and difference of cubes, evaluate $f(4)$.

a. 19

b. 22

c. 12

d. 23

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Factoring with sum and difference of cubes • Cross-Lesson: 1.01 Functions & Their Graphs

Find the positive solutions to the quadratic equation $x^2 - 11x + 24 = 0$.

a. $x = 3$ and $x = 9$

b. $x = 3$ and $x = 8$

c. $x = -3$ and $x = -8$

d. $x = 11$ and $x = 24$

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39 [Challenging] • Target: Factoring with sum and difference of cubes • Cross-Lesson: 10.01 Matrices & Systems

Find the positive solutions to the quadratic equation $x^2 - 13x + 36 = 0$.

a. $x = 4$ and $x = 10$

b. $x = 4$ and $x = 9$

c. $x = -4$ and $x = -9$

d. $x = 13$ and $x = 36$

40 [Challenging] • Target: Factoring polynomials completely • Cross-Lesson: 11.01 Probability & Statistics

Find the positive solutions to the quadratic equation $x^2 - 4x + 3 = 0$.

a. $x = 1$ and $x = 4$

b. $x = 1$ and $x = 3$

c. $x = -1$ and $x = -3$

d. $x = 4$ and $x = 3$

41 [Challenging] • Target: Factoring polynomials completely • Cross-Lesson: 3.01 Polynomial Functions

For the model $f(x) = 4x + 8$ in factoring polynomials completely, evaluate $f(3)$.

a. 20

b. 24

c. 12

d. 24

42 [Challenging] • Target: Factoring polynomials completely • Cross-Lesson: 4.01 Rational Expressions

For the model $f(x) = 4x + 8$ in factoring polynomials completely, evaluate $f(4)$.

a. 24

b. 28

c. 16

d. 28

43 [Challenging] • Target: Factoring polynomials completely • Cross-Lesson: 5.01 Exponential & Logarithmic Functions

Find the positive solutions to the quadratic equation $x^2 - 8x + 15 = 0$.

a. $x = 3$ and $x = 6$

b. $x = 3$ and $x = 5$

c. $x = -3$ and $x = -5$

d. $x = 8$ and $x = 15$

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44 [Challenging] • Target: Review: Graphs and characteristics of quadratic functions • Cross-Lesson: 6.01 Radicals & Rational Exponents

A data set recorded for review: graphs and characteristics of quadratic functions consists of the four values $\{14, 18, 22, 28\}$. What is the arithmetic mean?

a. 18.5

b. 19.5

c. 20.5

d. 22.5

45 [Challenging] • Target: Review: Solving quadratic equations by factoring • Cross-Lesson: 7.01 Sequences & Series

A data set recorded for review: solving quadratic equations by factoring consists of the four values $\{15, 19, 23, 29\}$. What is the arithmetic mean?

a. 19.5

b. 20.5

c. 21.5

d. 23.5

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